

Introduction to Artificial Intelligence

Lecture 4: Quantifying uncertainty

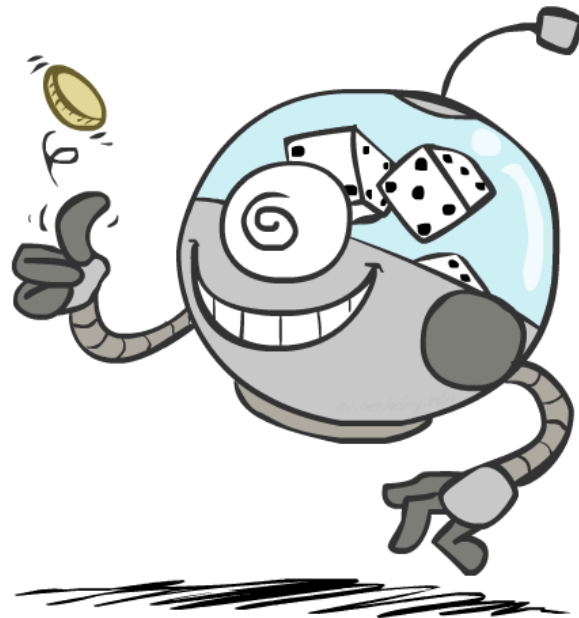
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Today

- Random variables
- Probability distributions
- Inference
 - by enumeration
 - with independence
 - with the Bayes' rule



Do not overlook this lecture!



Principle of maximum expected utility

From Lecture 1, a rational agent chooses the action of **highest expected utility**, given what it has perceived,

$$a_t = \arg \max_{a \in \mathcal{A}} \mathbb{E} [V(s_{1:T}) \mid e_{1:t}, a_t = a],$$

where $V : \mathcal{S}^* \rightarrow \mathbb{R}$ scores the states $s_{1:T}$, and the average is over the states the agent may reach, following $P(s_{t+1} \mid s_t, a_t)$ and $P(e_t \mid s_t)$.

What are these probabilities, and how do we reason with them?

Quantifying uncertainty

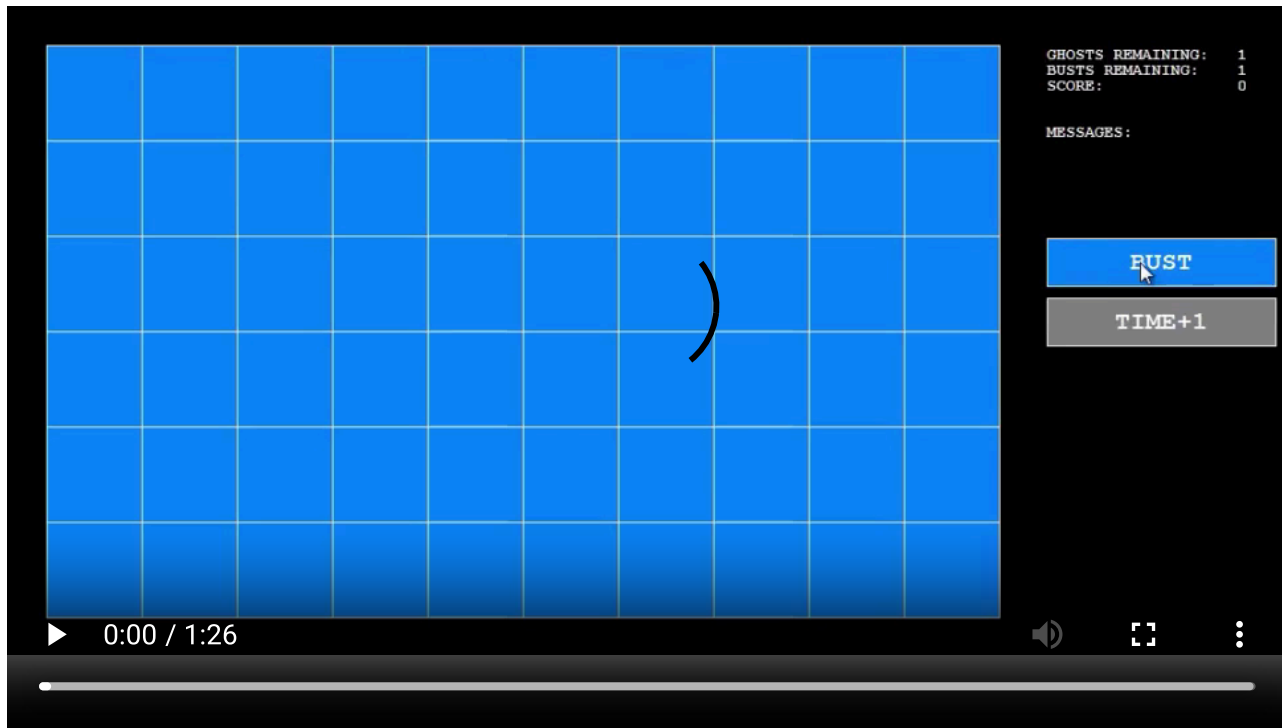
A ghost is **hidden** in the grid somewhere.

Sensor readings tell how close a square is to the ghost:

- On the ghost: red
- 1 or 2 away: orange
- 3 away: yellow
- 4+ away green



Sensors are **noisy**, but we know the probability values $P(\text{color} \mid \text{distance})$, for all colors and all distances.



Uncertainty

General setup:

- **Observed** variables or evidence: agent knows certain things about the state of the world (e.g., sensor readings).
- **Unobserved** variables: agent needs to reason about other aspects that are uncertain (e.g., where the ghost is).
- (Probabilistic) **model**: agent knows or believes something about how the observed variables relate to the unobserved variables.

Probabilistic reasoning provides a framework for managing our knowledge and beliefs.

Probabilistic assertions

A **probabilistic assertion** attaches a number to a proposition,

$$P(\text{ghost in cell } [3, 2]) = 0.02,$$

as a degree of belief, not as a claim about the truth of the proposition.

The agent holds beliefs rather than certainties for two reasons:

- **ignorance**: it does not observe the state, and its models of the world are approximate;
- **laziness**: listing every case and every exception would cost more than it is worth.

Frequentism vs. Bayesianism

What does a probability value represent? Take $P(\text{ghost in cell } [3, 2]) = 0.02$.

- The objectivist, **frequentist** view: a property of the world, the fraction of games in which the ghost sits in that cell.
- The subjectivist, **Bayesian** view: the degree of belief of the agent that the ghost is in that cell, given the readings it has taken.

The agent plays a single game, with no long run to count, so we read probabilities as beliefs in this course: a new reading changes the number, not the ghost.

Kolmogorov's axioms

Begin with a set Ω , the **sample space**. Its elements $\omega \in \Omega$ are the **sample points**, or possible worlds.

A **probability space** is a sample space equipped with a probability function P that assigns a number $P(\omega)$ to each sample point, such that

- $0 \leq P(\omega) \leq 1$ for all $\omega \in \Omega$;
- $\sum_{\omega \in \Omega} P(\omega) = 1$.

Events

An **event** is a set of sample points $A \subseteq \Omega$, and its probability is the mass it collects,

$$P(A) = \sum_{\omega \in A} P(\omega).$$

Equivalently, in Kolmogorov's terms,

- $P(A) \geq 0$ for all events;
- $P(\Omega) = 1$;
- $P(A \cup B) = P(A) + P(B)$ for disjoint events.

It follows that

$$P(\neg a) = 1 - P(a), \quad P(a \vee b) = P(a) + P(b) - P(a \wedge b).$$

Example

- Ω = the 6 possible rolls of a die.
- ω_i (for $i = 1, \dots, 6$) are the sample points, each corresponding to an outcome of the die.
- Assignment P for a fair die:

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$$

- "3 or more" is the event $A = \{3, 4, 5, 6\}$, of probability

$$P(A) = P(3) + P(4) + P(5) + P(6) = \frac{2}{3},$$

hence $P(\neg A) = 1 - P(A) = \frac{1}{3}$ for "2 or less".

Random variables

- A **random variable** is a function $X : \Omega \rightarrow D_X$ from the sample space to a domain D_X of outcomes.
- " $X = x$ " is the event $\{\omega \in \Omega : X(\omega) = x\}$, so P induces a **probability distribution** over X ,

$$P(X = x) = \sum_{\{\omega: X(\omega)=x\}} P(\omega).$$

- The events we care about are the **assignments** of random variables, such as $X_1 = x_1, \dots, X_n = x_n$.

Example:

On the die, "odd" is the random variable

$$\text{Odd} : \Omega \rightarrow \{\text{true}, \text{false}\}, \quad \text{Odd}(\omega) = (\omega \bmod 2 = 1),$$

whose distribution is

$$P(\text{Odd} = \text{true}) = P(1) + P(3) + P(5) = \frac{1}{2}, \quad P(\text{Odd} = \text{false}) = \frac{1}{2}.$$

Notations

- Random variables are named in upper case, X , Y , $Weather$; their values in lower case, x , y , sun .
- D_X is the domain of X , so that $x \in D_X$ is one of its values.
- $P(X = x)$ is a **number**, abbreviated $P(x)$ when clear from context.
- $\mathbf{P}(X)$ is the **distribution** of X , i.e. the table of the numbers $P(X = x)$ for all $x \in D_X$, and not a single value.

Probability distributions

Probability mass function

For a discrete variable X , the probability distribution $\mathbf{P}(X)$ is the list of the values $P(X = x)$ for all $x \in D_X$, known as the **probability mass function**.

$\mathbf{P}(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0



Joint distributions

A **joint** probability distribution over a set of random variables $X_1, \dots, X_n$ specifies the probability of each (combined) outcome:

$$P(X_1 = x_1, \dots, X_n = x_n) = \sum_{\{\omega: X_1(\omega)=x_1, \dots, X_n(\omega)=x_n\}} P(\omega)$$

$$\mathbf{P}(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Over n boolean variables, the table has 2^n entries, hence $2^n - 1$ free numbers.

Marginal distributions

The **marginal distribution** of a subset of random variables is obtained by summing the joint distribution over the values of the other variables (the **sum rule**),

$$P(x) = \sum_y P(x, y), \quad \mathbf{P}(X) = \sum_y \mathbf{P}(X, Y = y).$$

$\mathbf{P}(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$\mathbf{P}(T)$

T	P
hot	0.5
cold	0.5

$\mathbf{P}(W)$

W	P
sun	0.6
rain	0.4

$$P(t) = \sum_w P(t, w)$$

$$P(w) = \sum_t P(t, w)$$

Intuitively, marginal distributions are sub-tables which eliminate variables.

Conditional distributions

The **conditional probability** of a given b , for $P(b) > 0$, is the ratio of the probability of a and b to the probability of b ,

$$P(a \mid b) = \frac{P(a, b)}{P(b)}.$$

Indeed, observing $B = b$ rules out all those possible worlds where $B \neq b$, leaving a set whose total probability is just $P(b)$. Within that set, the worlds for which $A = a$ satisfy $A = a \wedge B = b$ and constitute a fraction $P(a, b)/P(b)$.



Conditional distributions are probability distributions over some variables, given **fixed** values for others. Writing $\mathbf{P}(X \mid Y)$ denotes them all at once: one distribution $\mathbf{P}(X \mid Y = y)$ for each value y of Y .

$$\mathbf{P}(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$\mathbf{P}(W \mid T = \text{hot})$$

W	P
sun	0.8
rain	0.2

$$\mathbf{P}(W \mid T = \text{cold})$$

W	P
sun	0.4
rain	0.6

Product rule

Rearranging the definition of the conditional probability,

$$P(a, b) = P(a \mid b)P(b), \quad \mathbf{P}(A, B) = \mathbf{P}(A \mid B)\mathbf{P}(B),$$

the second form standing for one equation per value of A and B .

Summing it over the values of B gives the **law of total probability**,

$$P(a) = \sum_b P(a \mid b)P(b).$$

$\mathbf{P}(W)$

W	P
sun	0.8
rain	0.2

$\mathbf{P}(D \mid W)$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3

$\mathbf{P}(D, W)$

D	W	P
wet	sun	?
dry	sun	?
wet	rain	?
dry	rain	?

Chain rule

More generally, any joint distribution can always be written as an incremental product of conditional distributions:

$$P(x_1, x_2, x_3) = P(x_1)P(x_2 \mid x_1)P(x_3 \mid x_1, x_2)$$
$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i \mid x_1, \dots, x_{i-1})$$

In the same way, $\mathbf{P}(X_1, \dots, X_n) = \prod_{i=1}^n \mathbf{P}(X_i \mid X_1, \dots, X_{i-1})$.

Inference

What an agent asks, given what it has observed:

$$P(\text{on time} \mid \text{no reported accidents}) = 0.9$$

$$P(\text{on time} \mid \text{no reported accidents, 5AM}) = 0.95$$

$$P(\text{on time} \mid \text{no reported accidents, rain}) = 0.8$$

$$P(\text{ghost in } [3, 2] \mid \text{red in } [3, 2]) = 0.99$$

Probabilistic inference

- **Evidence** variables $\mathbf{E} = \{E_1, \dots, E_k\}$, observed as $\mathbf{e} = (e_1, \dots, e_k)$.
- **Query** variable Q .
- **Hidden** variables $\mathbf{H} = \{H_1, \dots, H_r\}$.
- Together, they are all the variables of the model,
 $\{Q\} \cup \mathbf{E} \cup \mathbf{H} = \{X_1, \dots, X_n\}$.

Inference is the problem of computing the posterior distribution $\mathbf{P}(Q \mid \mathbf{e})$.

Normalization trick

$$\mathbf{P}(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$\rightarrow \mathbf{P}(T = \text{cold}, W)$$

T	W	P
cold	sun	0.2
cold	rain	0.3

Select the joint probabilities matching the evidence $T = \text{cold}$.

$$\rightarrow \mathbf{P}(W \mid T = \text{cold})$$

W	P
sun	0.4
rain	0.6

Normalize the selection (make it sum to **1**).

Inference by enumeration

Starting from the joint distribution $\mathbf{P}(Q, \mathbf{E}, \mathbf{H})$, select the entries consistent with the evidence, sum the hidden variables out, and normalize:

$$\mathbf{P}(Q \mid \mathbf{e}) = \frac{1}{Z} \sum_{\mathbf{h}} \mathbf{P}(Q, \mathbf{h}, \mathbf{e}), \quad Z = \sum_q \sum_{\mathbf{h}} P(q, \mathbf{h}, \mathbf{e}).$$

function ENUMERATION-ASK(Q, e, P) **returns** a distribution over Q

$b \leftarrow$ an empty table over the values of Q

for each value q of Q **do**

$b[q] \leftarrow$ the sum of $P(q, h, e)$ over the assignments h

$Z \leftarrow$ the sum of $b[q]$ over the values q

return b / Z

Example:

- $\mathbf{P}(W)$?
- $\mathbf{P}(W \mid \text{winter})$?
- $\mathbf{P}(W \mid \text{winter, hot})$?

S	T	W	P
summer	hot	sun	0.3
summer	hot	rain	0.05
summer	cold	sun	0.1
summer	cold	rain	0.05
winter	hot	sun	0.1
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.2

- Inference by enumeration can be used to answer probabilistic queries for **discrete variables** (i.e., with a finite number of values).
- However, enumeration **does not scale!**
 - Assume a domain described by n variables taking at most d values.
 - Space complexity: $O(d^n)$
 - Time complexity: $O(d^n)$

Can we reduce the size of the representation of the joint distribution?

The joint distribution has d^n entries, but the chain rule already writes it as a product of conditional distributions,

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i \mid x_1, \dots, x_{i-1}).$$

Independence and **conditional independence** state that some of these factors do not depend on all their conditions. Each dropped variable divides the size of a factor by d , which is what makes probabilistic models **scale up**.

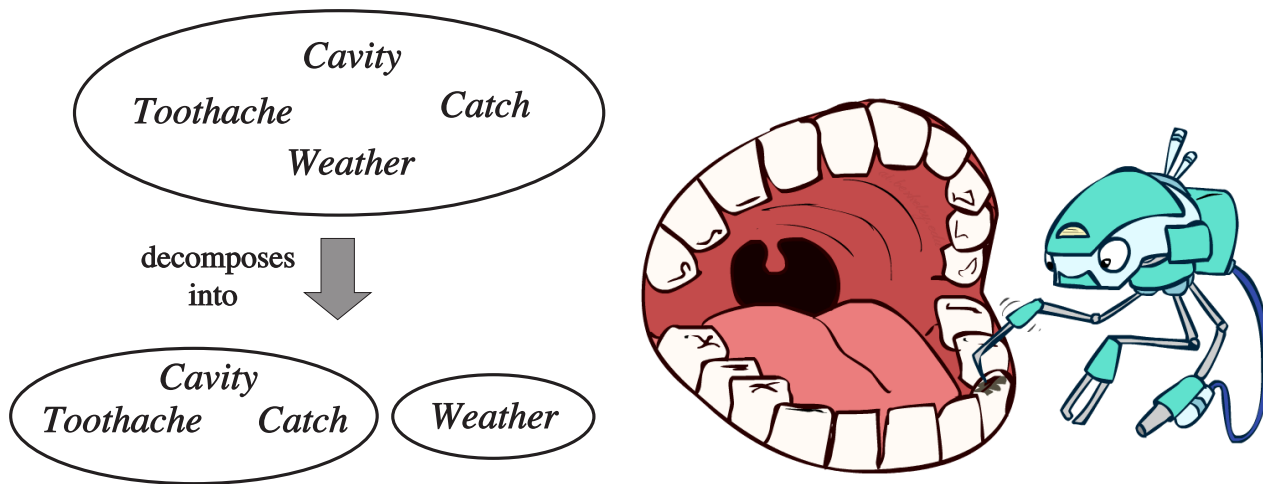
Independence

A and B are **independent**, denoted $A \perp B$, iff

$$P(a, b) = P(a)P(b) \quad \text{for all } a \in D_A, b \in D_B.$$

Equivalently, knowing one says nothing about the other: $P(a \mid b) = P(a)$ whenever $P(b) > 0$, and $P(b \mid a) = P(b)$ whenever $P(a) > 0$.

Example 1:



$$\begin{aligned} &P(\text{toothache}, \text{catch}, \text{cavity}, \text{weather}) \\ &= P(\text{toothache}, \text{catch}, \text{cavity})P(\text{weather}) \end{aligned}$$

The original 32-entry table reduces to one 8-entry and one 4-entry table (assuming 4 values for **Weather** and boolean values otherwise).

Example 2:

For n independent coin flips, the joint distribution can be fully **factored** and represented as the product of n 1-entry tables.

- $2^n \rightarrow n$

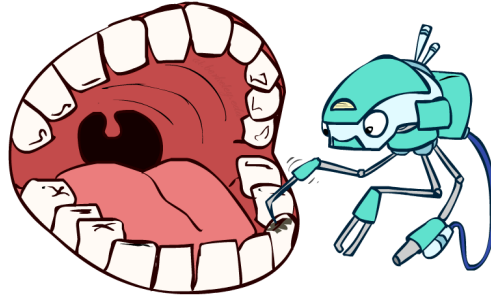
Conditional independence

A and B are **conditionally independent** given C , denoted $A \perp B \mid C$, iff

$$P(a, b \mid c) = P(a \mid c)P(b \mid c) \quad \text{for all } a \in D_A, b \in D_B, c \in D_C \text{ with } P(c) > 0.$$

Equivalently, once c is known, one says nothing more about the other:

$$P(a \mid b, c) = P(a \mid c) \text{ and } P(b \mid a, c) = P(b \mid c), \text{ whenever these are defined.}$$



Example 1:

Assume three random variables **Toothache**, **Catch** and **Cavity**.

Catch is conditionally independent of **Toothache**, given **Cavity**. Therefore, we can write:

$$\begin{aligned} P(\text{toothache}, \text{catch}, \text{cavity}) \\ &= P(\text{toothache} \mid \text{catch}, \text{cavity})P(\text{catch} \mid \text{cavity})P(\text{cavity}) \\ &= P(\text{toothache} \mid \text{cavity})P(\text{catch} \mid \text{cavity})P(\text{cavity}) \end{aligned}$$

In this case, the representation of the joint distribution reduces to $2 + 2 + 1$ independent numbers (instead of $2^n - 1$).

Example 2 (Naive Bayes):

More generally, from the product rule, we have

$$P(\text{cause}, \text{effect}_1, \dots, \text{effect}_n) = P(\text{effect}_1, \dots, \text{effect}_n \mid \text{cause})P(\text{cause})$$

Assuming the effects are **conditionally independent given the cause**, it comes:

$$P(\text{cause}, \text{effect}_1, \dots, \text{effect}_n) = P(\text{cause}) \prod_i P(\text{effect}_i \mid \text{cause})$$

This probabilistic model is called a **naive Bayes** model.

- The complexity of this model is $O(n)$ instead of $O(2^n)$ without the conditional independence assumptions.
- Naive Bayes can work surprisingly well in practice, even when the assumptions are wrong.

Causal and diagnostic directions

A model is usually given in the **causal** direction: a prior on the cause, and the distribution of its effects given the cause, as in the naive Bayes model. For the ghost and its sensors, $\mathbf{P}(G)$ and $\mathbf{P}(R_{i,j} \mid G)$.

Inference asks the **diagnostic** question, $\mathbf{P}(G \mid r_{i,j})$: the cause, given the effects that were observed.

Bayes' rule turns one into the other, and follows from the two factorizations of the product rule.

Study the next slide. **Twice.**

The Bayes' rule

The product rule defines two ways to factor the joint distribution of two random variables.

$$P(a, b) = P(a | b)P(b) = P(b | a)P(a)$$

Therefore,

$$P(a | b) = \frac{P(b | a)P(a)}{P(b)}.$$



- $P(a)$ is the prior belief on a .
- $P(b)$ is the probability of the evidence b .
- $P(a | b)$ is the posterior belief on a , given the evidence b .
- $P(b | a)$ is the conditional probability of b given a . Depending on the context, this term is called the likelihood.

$$P(a \mid b) = \frac{P(b \mid a)P(a)}{P(b)}$$



The Bayes' rule is the **foundation** of many AI systems.

Example 1: diagnostic probability from causal probability.

$$P(\text{cause} \mid \text{effect}) = \frac{P(\text{effect} \mid \text{cause})P(\text{cause})}{P(\text{effect})}$$

where

- $P(\text{effect} \mid \text{cause})$ quantifies the relationship in the **causal** direction.
- $P(\text{cause} \mid \text{effect})$ describes the **diagnostic** direction.

Let S =stiff neck and M =meningitis. Given $P(s \mid m) = 0.7$, $P(m) = 1/50000$, $P(s) = 0.01$, it comes

$$P(m \mid s) = \frac{P(s \mid m)P(m)}{P(s)} = \frac{0.7 \times 1/50000}{0.01} = 0.0014.$$

Example 2: Ghostbusters, revisited

- A random variable G gives the ghost location, and a random variable $R_{i,j}$ the color read at the cell $[i, j]$.
- We start from a uniform **prior** $\mathbf{P}(G)$ over the locations.
- We know the sensors, i.e. the **sensor model** $\mathbf{P}(R_{i,j} \mid G)$.
 - e.g., $P(R_{1,1} = \text{yellow} \mid G = [1, 1]) = 0.1$.
- Readings are conditionally independent given the ghost location,
 $R_{i,j} \perp R_{i',j'} \mid G$.

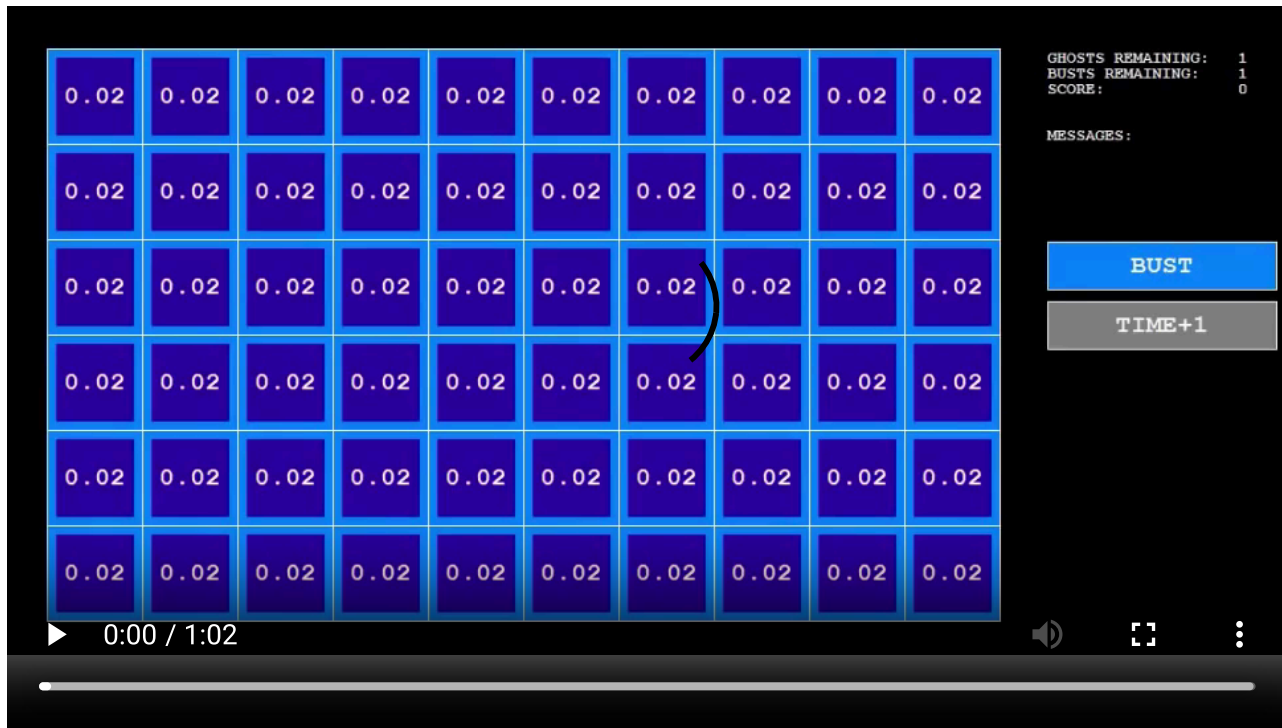
Once the reading r_1 is observed, Bayes' rule gives the **posterior** over the locations,

$$\mathbf{P}(G \mid r_1) = \frac{\mathbf{P}(r_1 \mid G)\mathbf{P}(G)}{P(r_1)} = \frac{1}{Z}\mathbf{P}(r_1 \mid G)\mathbf{P}(G).$$

Each new reading updates the posterior of the previous ones, which acts as its prior,

$$\mathbf{P}(G \mid r_{1:k+1}) = \frac{1}{Z}\mathbf{P}(r_{k+1} \mid G)\mathbf{P}(G \mid r_{1:k}),$$

since the readings are conditionally independent given G .



Summary

- Uncertainty is **inescapable**, and **probability theory** is the framework we use to represent it: numbers attached to propositions, measuring the agent's beliefs rather than the world.
- The joint distribution answers every query: select, marginalize, normalize. It costs $O(d^n)$.
- **Independence** and **conditional independence** factor it into smaller tables. Naive Bayes goes from $O(2^n)$ down to $O(n)$.
- **Bayes' rule** turns a causal model into a diagnostic belief, and each posterior is the prior of the next observation.

All the rules

Event	$P(A) = \sum_{\omega \in A} P(\omega)$
Complement	$P(\neg a) = 1 - P(a)$
Union	$P(a \vee b) = P(a) + P(b) - P(a \wedge b)$
Sum rule	$P(x) = \sum_y P(x, y)$
Conditioning	$P(a \mid b) = P(a, b) / P(b)$
Product rule	$P(a, b) = P(a \mid b)P(b) = P(b \mid a)P(a)$
Total probability	$P(x) = \sum_y P(x \mid y)P(y)$
Chain rule	$P(x_{1:n}) = \prod_i P(x_i \mid x_{1:i-1})$
Bayes' rule	$P(a \mid b) = P(b \mid a)P(a) / P(b)$
Enumeration	$\mathbf{P}(Q \mid \mathbf{e}) = \frac{1}{Z} \sum_{\mathbf{h}} \mathbf{P}(Q, \mathbf{h}, \mathbf{e})$
Independence, $X \perp Y$	$P(x, y) = P(x)P(y)$
Conditional independence, $X \perp Y \mid Z$	$P(x, y \mid z) = P(x \mid z)P(y \mid z)$
Naive Bayes	$P(c, e_{1:n}) = P(c) \prod_i P(e_i \mid c)$

The end.